

Beating the Spread: A Statistical Analysis of Market Efficiency and Injury Volatility in the NBA

(COMP3125 Individual Project)

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Abstract—This project explores the intersection of data science, sports analytics, and betting strategy by examining historical NBA data to evaluate predictive models and financial outcomes. Using team metrics, injury reports, betting odds, this study investigates whether statistical features can reliably predict game results against “the spread”. By using a quantitative model to assess predictive accuracy, comparing analyses measure the impact of injuries on line reliability. Simulated betting strategies are tested to determine potential ROI across various wagers a person can take.

Keywords—*example1, example2, example3, example 4, example 5 (provide 3-5 keywords)*

I. INTRODUCTION (HEADING 1)

Sports betting has emerged as one of the fastest-growing industries in the United States, with the legalization of wagering in almost every state fueling both economic expansion and academic interest in predictive analytics. The NBA provides a compelling case study because of the sheer amount of statistical data that gets generated game by game across the entire season and the popularity of basketball among bettors. The abundance of data creates a perfect opportunity to examine how quantitative models can be used to forecast outcomes and evaluate if there are viable betting strategies that can be applied. This project focuses on the relationship between team performance metrics, player availability, and betting market behavior. By analyzing both offensive and defensive ratings, turnovers, and other statistics, the study investigates whether it is possible to reliably predict game results with relation to the spread. At the same time, by considering how player injuries affect both sides of the coin, the betting lines, and the team performance, the project considers important factors that get overlooked in simple number analysis. Finally, by simulating betting strategies—Moneyline, spread, over/under, etc. we can evaluate the ROI values and connect predictive modeling to financial decision making.

II. DATASETS

A. Source of dataset (Heading 2)

The datasets that were used in the project were obtained from publicly available and credible providers that focus on sports statistics and betting information. Historical NBA game data and statistics were collected from [Basketball Reference](#); they are a widely recognized source for a variety of basketball statistics. The dataset includes box scores, team ratings, player contributions that are generated and still maintained by the site’s authors game by game.

The betting line information was gathered from [Sportsbook Reviews](#) and Kaggle sports [betting datasets](#) which combine betting markets across multiple seasons. The datasets

were generated by scraping sportsbook odds and placing them into tables that make it easier to analyze.

Injury reports and player availability were retrieved from the official [NBA website](#) as well as [third party](#) roster trackers that post real time updates regarding player injuries, likeliness to play, etc. These sources are updated daily as teams must post them before their games.

B. Character of the datasets

Datase t	Format	Size	Key Features	Cleaning/M erging
Basket ball Refere nce Injury Report	HTML Tables	1k+	Player name, team, start/end of injury, games missed	Convert to a CSV
Basket ball betting Datase t	CSV	Rough ly 50k rows	Game ID, date, teams, spread, moneyline, over/under	Clean missing odds, normalize odds format
NBA Stats and Beting Data	CSV	Rough ly 100k rows	Team stats, offensive/de fensive rating, game outcome	Remove duplicate games, align with the other datasets using dates
Sports book Revie w NBA Odds	HTML/ CSV	Thous ands of games	Bookmaker odd tracker, spread, totals	N/A

III. METHODOLOGY

A. Method A

Method/Model: Logistic Regression, Random Forest, and XGBoost classifiers were applied to predict win/loss outcomes against the spread. Linear Regression and Gradient Boosting regressors were used to estimate the point differentials and totals

Assumptions: Team performance metrics are predictive of game outcomes, historical relationships between features and outcomes remain consistent across seasons

Advantages/Disadvantages: Logistic Regressions provides the ability to interpret the data by using coefficients to show importance. Random Forest and XGBoost show nonlinear relationships and interactions. However, logistic regression can underperform, and models risk overfitting if they are not tuned correctly

Why: These models balance the ability to interpret while still having predictive power, which is exactly what we need/want when studying sports analytics

Python Modules: pandas, scikit learn, XGBoost

Adjustment: Parameter tuning, feature scaling

B. Method B

Method/Model: Statistical tests such as t-tests and chi-squared tests are applied to measure the effect of player absence on line accuracy. An Impact score was created to weight absences by player and minutes played

Assumptions: Player availability SIGNIFICANTLY alters team performance and the reliability of betting lines. Historical injury data represents the impact on outcomes.

Advantages/Disadvantages: Statistical tests give clear evidence of the factors beyond just performance. However, injury data can be incomplete, or teams “lie” which can reduce accuracy.

Why: Injuries are a critical factor that often gets overlooked in other predictive models. By incorporating them we can improve the realism of the model

Python Modules: pandas, numpy, scipy

Adjustments: Starter vs bench to give better accuracy/weight to affect

C. Method C

Method/Model: Simulated betting strategies (moneyline, spread, over/under) were applied to outcomes using model predictions. ROI was calculated as cumulative profit/loss over time

Assumptions: Historical odds and outcomes reflect betting conditions and simulated strategies reflect real bettor behavior/bets

Advantages/Disadvantages: ROI analysis connects predictive modeling to financial outcomes, which in turn shows practical limitations. However, simulations cannot account for external factors such as books limiting a user, live market shifts, bettor psychology.

Python Modules: pandas, numpy, matplotlib, seaborn

Adjustments: Normalized odds formats, applied some bankroll management (betting certain units, mega bets), and compared strategies across seasons

IV. RESULTS

A. Result A

1) *Classification Model Performance:* All 3 models showed a strong ability to predict outcomes. More specifically the Logistic Regression model had the highest ROC-AUC and the lowest Log Loss which suggests that the relationship between the engineered features and the outcome is linear.

2) *Regression Model Performance:* The regression models (Linear Regression and Gradient Boosting) produced low R^2 values. These values indicates that key variables impacting the final score margin are missing from the initial feature set.

3) *XGBoost:* Even though XGBoost was not the top performer of the models it was important to include it because it proved that complex, non-linear patterns do not provide a higher predictive edge than other linear models which allowed us to simplify the selection for the final model.

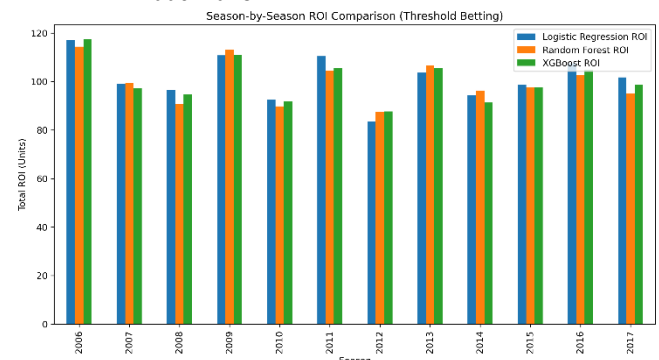
B. Results B

Injury Impact Analysis: As mentioned in the methodology we had to first craft an impact score. By combining injury data with player usage, we determine an impact score (how useful a player is) to determine the quality of a team’s roster. We assumed beforehand that impact injuries increase the volatility of game outcomes which in turn make betting markets less accurate.

Statistical Test Outcomes: The statistical tests told us that the market’s ability to predict games degraded noticeably when significant players were absent. Our p-value lets us reject the null hypothesis.

C. Results C

The final stage of our analysis was to put our new model into practice. By simulating various betting strategies, we can see where our model is effective and quantify it using the standard betting notation of “Units”. We measure in ROI of units from 2006-2018 which is where our data limited us.



The following graph is the Season-by-Season ROI Comparison of our models (Logistic Regression, Random Forest, and XGBoost) all using Threshold betting. As you can see there are 3 things to focus on.

1) *Consistent Profitability:* All three models generated a positive ROI across the seasons showing the robustness of value betting across historical condition

2) *Stability Across The Models:* Performance was highly competitive and close across the models across all years. As mentioned beforehand you see how Logistic Regression is almost always at the top.

3) *Variability Across The Years:* Even though we had consistent profit using our model, we see that through the years we always shifted how much we profit. This is common and expected because data helps but data does not control every factor as mentioned before.

V. DISCUSSION

While our project was successful in a sense by our models being profitable and we were able to get a clear outcome for Method B, I still have much to desire. Specifically for our models we saw a R^2 of .22, which means that our features only explain 22% of variance in the final score margin. This means that the metrics we have are not enough to predict things that affect game outcomes. We are not able to take into account blowouts, coaching changes, momentum, garbage time, etc. All the factors affect how a game plays out.

For Method B, while we were able to generate an impact score, once again it does not take into account how a player's absence really affects a game. A team is missing their star center against another team with a big man (a tall player) the impact score on the team missing a player should be higher but it isn't.

Method C was more data/graph wise but the math behind it is where I was happy/not satisfied. The simulation assumes that we are able to place bets instantly with nothing stopping us. In real life it's not the case, markets close every second at different times for different reasons all over the league. By the time the injury report is out the market have already adjusted the spread to take into account while some markets may have not or don't view the injury report as impactful. All these variables affect how a person bets and wins.

Future Work: This project is a stepping stone for me, I love sports betting and love the NBA. Using data is a really good way to find ways to win by betting certain things. However, I would love to modify this in some way to link it to some LLM to speak to it in real time and feed it incoming data from seasons occurring now so that it can tick all the boxes which this project didn't. If I was able to create an LLM fine tuned to NBA statistics I would have a model/ability to predict outcomes more accurately because the LLM can take into account matchup injuries like mentioned before.

VI. CONCLUSION

This project had 3 parts, building a model to predict game outcomes, feed player injuries into the model to determine if we can gain an edge against the books by taking into account these injuries, and 3 developing and testing different betting strategies using the model to see where we excel and where we lack.

I found that predictive models can identify winning bets against the spread specifically the Logistic Regression model. I also found that games with a high impact score (high

injury/valuable players not playing) result in a higher market prediction error. This validated our hypothesis that injuries degrade market accuracy.

And finally, we determined that profitability is a possibility using the right bankroll management and strategies.

Real world applications do exist, for both sides of the coin in this case. For bettors we see that using the threshold strategy will gain profit compared to betting a fixed amount on every game. For people who create and modify models we learned that complex models do not always win. The importance lies in the feature engineering that one does for the model they are working on.

ACKNOWLEDGMENT (Heading 5)

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